

**JAMES RUSE AGRICULTURAL HIGH SCHOOL**  
**TERM 1 ASSESSMENT 1997**  
**YEAR 12 3/4 UNIT**

**Time: 85 minutes.**

**Hand in each question separately**

**QUESTION 1 ( 10 marks )**

- (a) After investing \$P for 6 years at a rate of 5% per annum compounded annually, the amount matured to \$268. Find P to the nearest dollar.
- (b) Differentiate with respect to x:

(i)  $y = \cos(x^2)$

(ii)  $y = \log_e(\sin 2x)$

- (c) Find the equation of the tangent to the curve  $y = \operatorname{cosec} x$  at  $x = \frac{\pi}{6}$ .

**QUESTION 2 ( 10 marks )**

- (a) Find the general solution to:  $\cos 2x - 1 = 0$ .
- (b) Use the table of integrals provided to find:

$$\int \left( \frac{\sec 4x \tan 4x}{4} \right) dx$$

- (c) Evaluate  $\cos^{-1}\left(-\frac{1}{2}\right)$  exactly (answer in radians).

- (d) Differentiate with respect to x:

$$y = \cos^{-1}\left(\frac{x}{4}\right)$$

- (e) Without the use of calculus, find the minimum value of  $(\tan^{-1}x + \sin x)$  for  $0 \leq x \leq 1$ . Justify your answer.

**QUESTION 3 ( 10 marks )**

- (a) Evaluate:
- (i)  $\int_0^2 \frac{dx}{4+x^2}$
- (ii)  $\int_0^2 \frac{4x}{4+x^2} dx$

- (b) (i) Draw a neat sketch of:  $y = \frac{\pi}{2} + 2 \sin^{-1}\left(\frac{x}{2}\right)$ .
- (ii) State its DOMAIN and RANGE.

**QUESTION 4 ( 10 marks )**

- (a) (i) Express  $(\sin x + \sqrt{3} \cos x)$  in the form  $A \cos(x - \alpha)$  for  $A > 0$  and  $\alpha$  acute.
- (ii) Find the maximum value of  $(\sin x + \sqrt{3} \cos x)$ .

- (b) Evaluate:  $\int_0^{\frac{\pi}{12}} (1 + \sin^2 2x) dx$

- (c) Given that  $f(x) = 2 - (x+1)^2$  is defined for  $x \geq -1$  and has an inverse  $f^{-1}(x)$ , neatly sketch the graph of  $y = f^{-1}(x)$ .

**QUESTION 5 ( 10 marks )**

- (a) Draw a neat sketch of  $y = \cos(\sin^{-1}x)$ , clearly showing all x,y intercepts.
- (b) John takes out a \$25,000 loan at 9% per annum, the interest compounded monthly, and agrees to pay the loan by equal monthly instalments over six years.
- (i) Show that each monthly repayment is approximately \$450.
- (ii) Assuming he repays \$450 per month, how much is still owed after making the 24th repayment?
- (iii) After the 24th repayment he makes a lump sum payment of \$10,000. If he wishes to still make 48 more repayments, what would be the new value of each repayment?

**QUESTION 6 ( 10 marks )**

- (a) The area bounded between the co-ordinate axes, the curve  $y = \frac{1}{\sqrt{1+4x^2}}$

and the line  $x = \frac{1}{2}$  is rotated about the x-axis.

- (i) Find the volume generated by the rotated area.
- (ii) Determine the volume as x increases indefinitely.
- (b) Given that  $y_1 = 2 \tan^{-1}\sqrt{x}$  and  $y_2 = \sin^{-1}\left(\frac{x-1}{x+1}\right)$  for  $x > 0$ ,
- (i) evaluate  $\frac{d}{dx}(y_1 - y_2)$ .
- (ii) If  $f(x) = y_1 - y_2$ , deduce that  $f(x)$  is a constant, and find the value of the constant.
- (iii) Draw a neat sketch of the function  $f(x)$ .

**END OF PAPER**

TERM 1 ASSESSMENT YEAR 12  
3/4 UNIT 1997 SOLUTIONS

1 (a)  $268 = P(1.05)^5$

$\therefore P = \$200$  2

(b)

(i)  $\frac{dy}{dx} = -2x \sin(x^2)$  2

(ii)  $\frac{dy}{dx} = \frac{1}{\sin 2x} \cdot 2 \cos 2x$   
 $= 2 \cot 2x$  2

(c)  $\frac{dy}{dx} = -\cot x \cot x$

$\frac{dy}{dx} = -\cot \frac{\pi}{6} \cot \frac{\pi}{6}$  at  $x = \frac{\pi}{6}$

Let  $m = -2\sqrt{3}$

Equation of tangent is:

$y - 2 = -2\sqrt{3}(x - \frac{\pi}{6})$   
 $= -2\sqrt{3}x + \frac{\sqrt{3}\pi}{3}$   
 $\therefore y = -2\sqrt{3}x + (\frac{\sqrt{3}\pi}{3} + 2)$  4

2 (a)  $\cos 2x = 1 = \cos 0$  (b)  $\frac{1}{16} \sec 4x + C$  (c)  $\pi - \cos^{-1}(\frac{1}{2})$

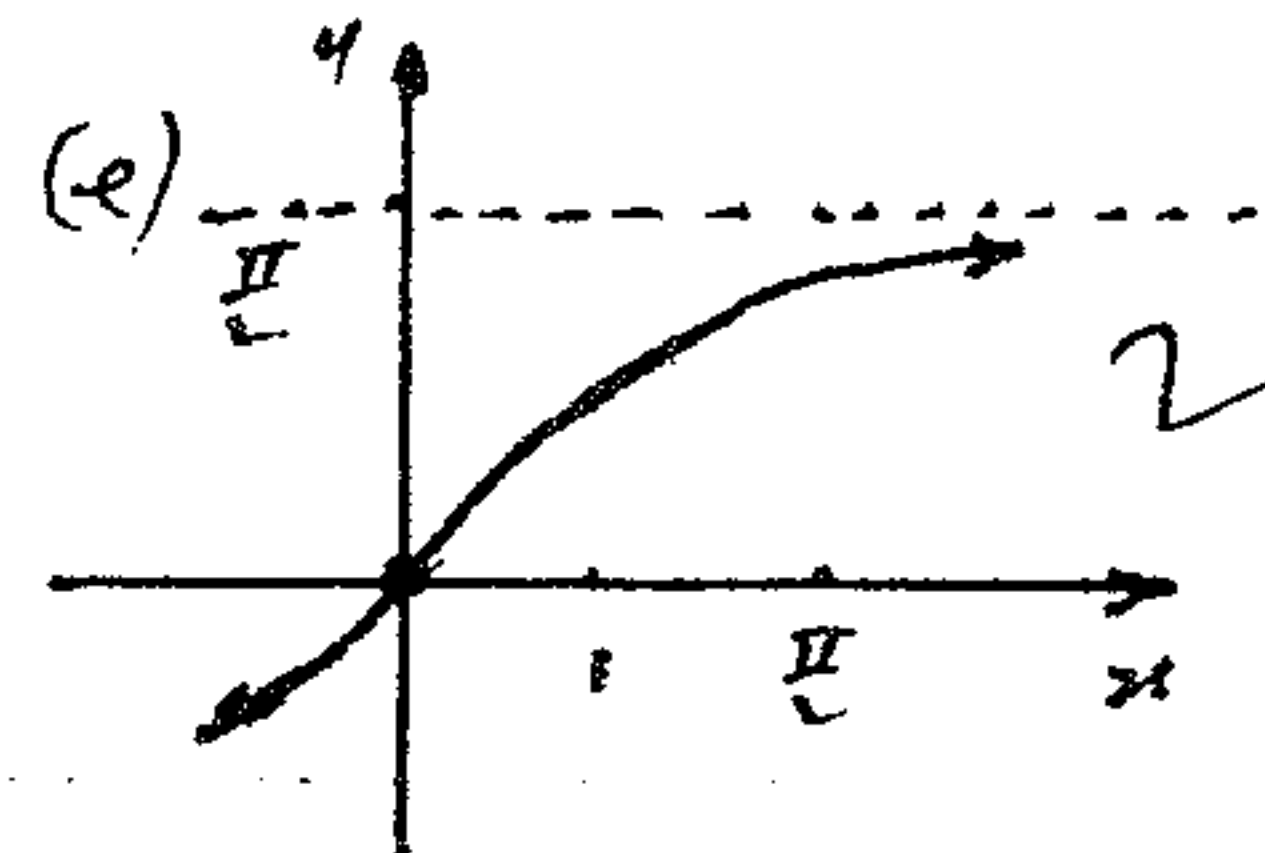
$\therefore 2x = 2n\pi$

$\therefore x = n\pi$  2

$\frac{1}{16} \sec 4x + C$

$= \pi - \frac{\pi}{3}$   
 $= \frac{2\pi}{3}$  2

(d)  $\frac{dy}{dx} = -\frac{1}{\sqrt{1-\frac{x^2}{16}}} \times \frac{1}{16}$   
 $= -\frac{1}{\sqrt{16-x^2}}$  2



Since both functions are increasing and  $f(0) = 1$   
 $\therefore$  MIN at  $x = 0$  10

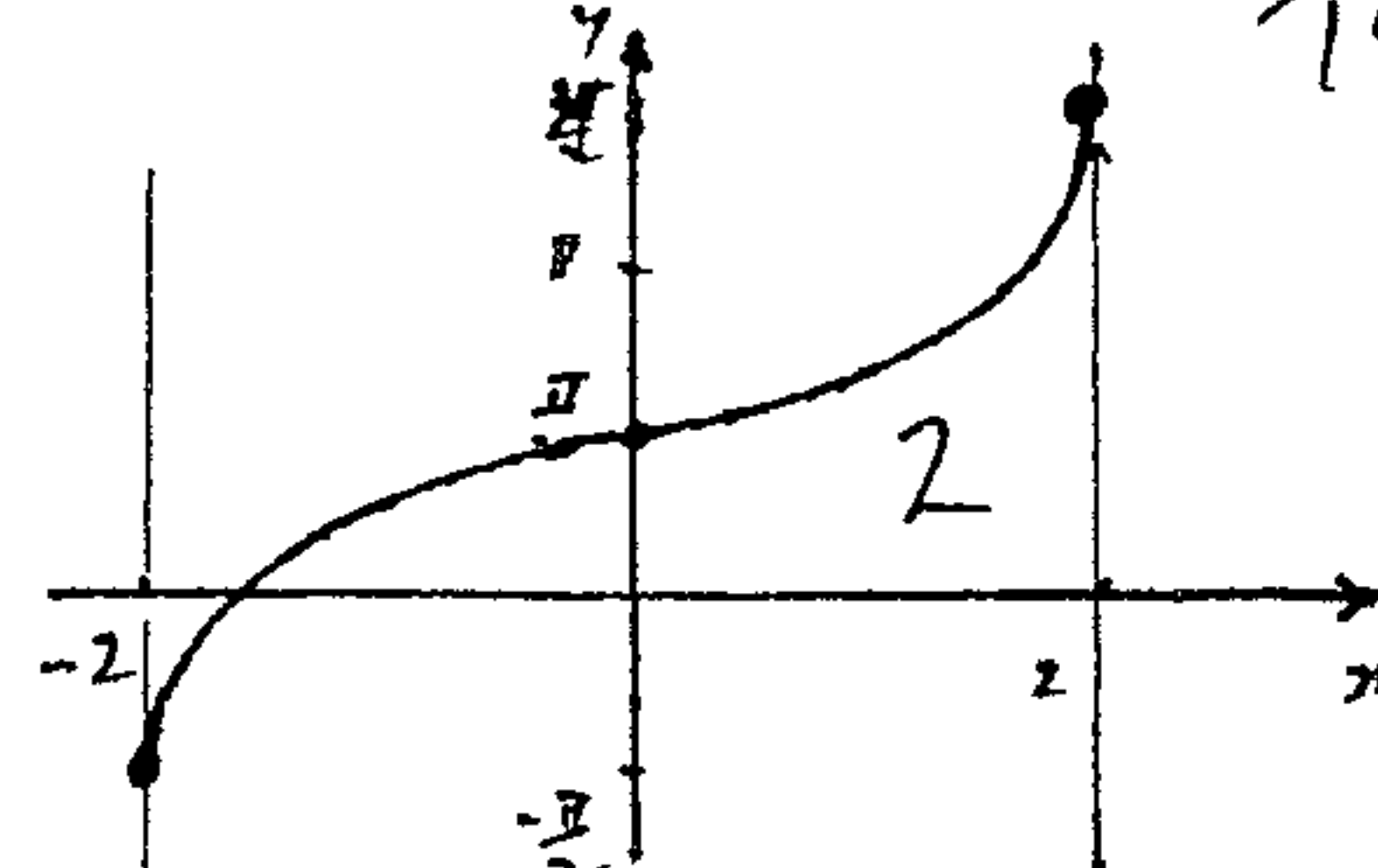
3 (a) (i)  $\int_0^2 \frac{dx}{4+x^2}$

$= \frac{1}{2} [\tan^{-1} \frac{x}{2}]_0^2$   
 $= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0)$   
 $= \frac{\pi}{8}$  3

(ii)  $\int_0^2 \frac{4x}{4+x^2} dx$

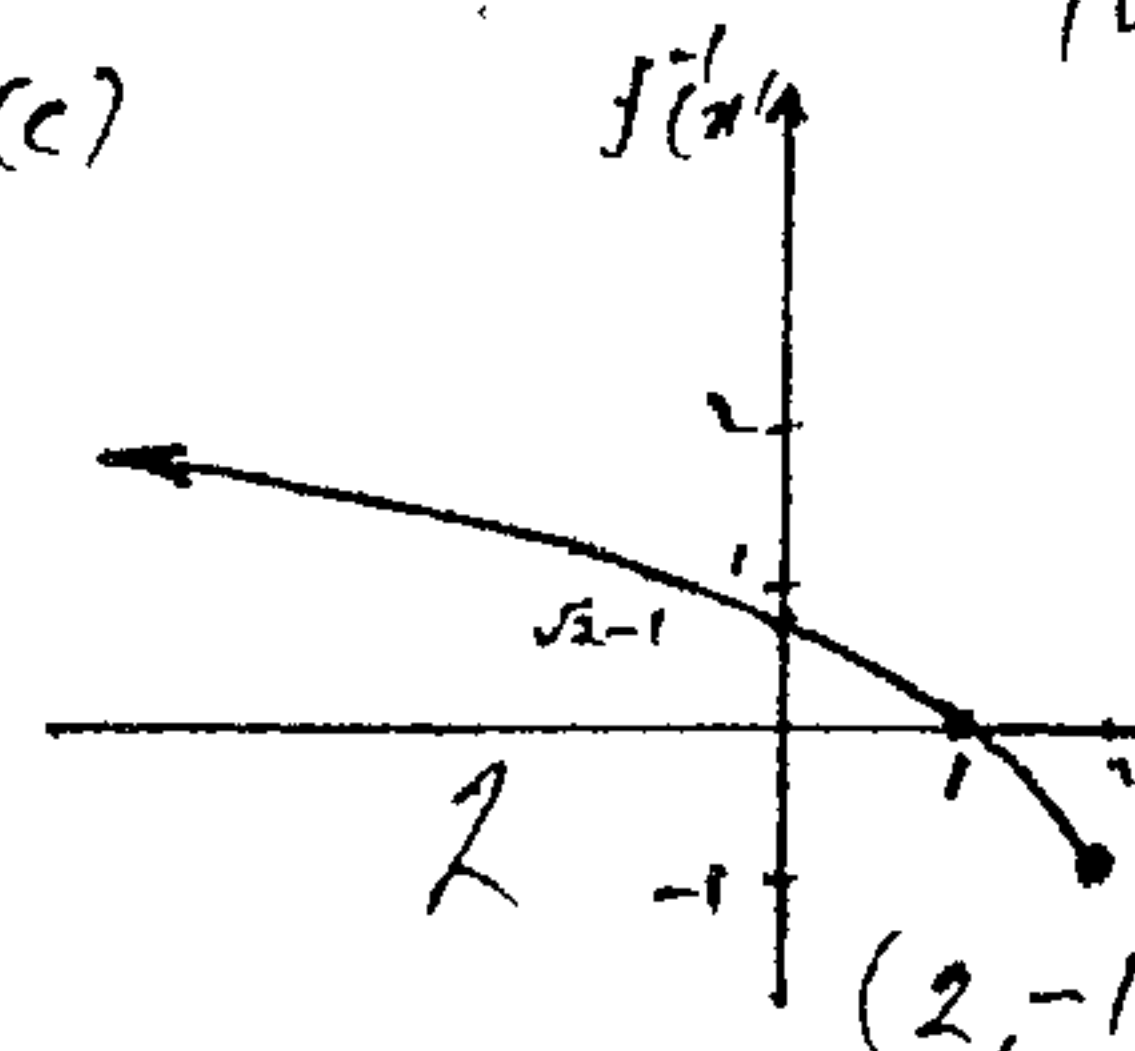
$= 2 [\ln(4+x^2)]_0^2$   
 $= 2 (\ln 8 - \ln 4)$   
 $= 2 \ln 2$  3

(b) (i)



$D = \{x: -2 \leq x \leq 2\}$   
 $R = \{y: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}\}$  10

(c)



$D = \{x: x \leq 2\}$  10

4 (a) (i)  $\sin x + \sqrt{3} \cos x$   
 $= A(\cos x \cos \alpha + \sin x \sin \alpha)$

$\Rightarrow A \cos \alpha = \sqrt{3}$   $A \sin \alpha = 1$

$\therefore \alpha = \frac{\pi}{6}$   $A = 2$

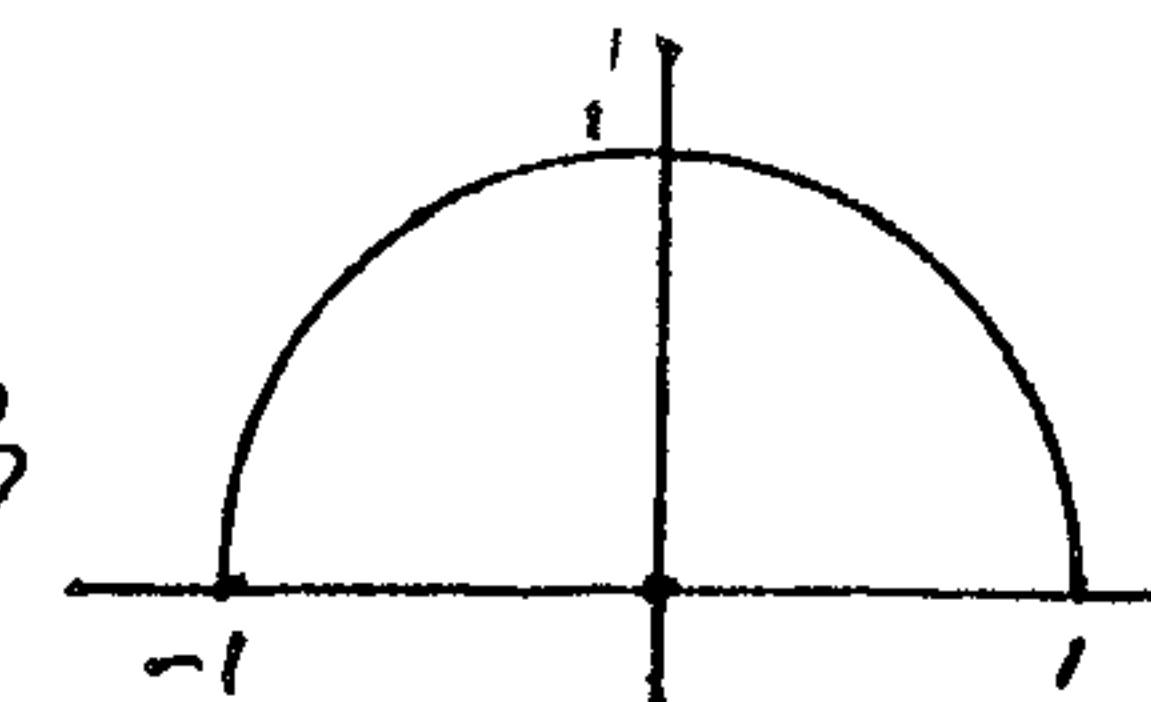
$\therefore \sin x + \sqrt{3} \cos x$   
 $= 2 \cos(x - \frac{\pi}{6})$  2

(ii) Max. value is 2

(b)  $\int_0^{\frac{\pi}{12}} (1 + \sin^2 2x) dx$

$= \int_0^{\frac{\pi}{12}} [1 + \frac{(1 - \cos 4x)}{2}] dx$   
 $= \frac{3}{2} \int_0^{\frac{\pi}{12}} dx - \frac{1}{2} \int_0^{\frac{\pi}{12}} \cos 4x dx$   
 $= \frac{3}{2} [x]_0^{\frac{\pi}{12}} - \frac{1}{8} [\sin 4x]_0^{\frac{\pi}{12}}$   
 $= \frac{3}{2} \cdot \frac{\pi}{12} - \frac{1}{8} (\sin \frac{\pi}{3} - 0)$   
 $= \frac{2\pi}{16} - \frac{\sqrt{3}}{16}$  4

5 (a)  $y = \cos(\sin^{-1} x)$  Let  $U = \sin^{-1} x$   $x = \sin U$   
 $y = \cos U$   $-1 \leq x \leq 1 \Rightarrow x^2 + y^2 = 1$   
 $0 \leq y \leq 1$   $-\frac{\pi}{2} \leq U \leq \frac{\pi}{2}$  for  $y \geq 0$



(b) (i) Let the amount be \$A.

$\frac{1}{25000} (\frac{403}{400})$   $\frac{25000 (\frac{403}{400})}{25000} - A$   
 $\frac{2}{25000 (\frac{403}{400})^2} - A (\frac{403}{400})$   $25000 (\frac{403}{400})^2 - A (\frac{403}{400}) - A$   
 $\frac{3}{25000 (\frac{403}{400})^3} - A (\frac{403}{400})^2 - A (\frac{403}{400}) - A$   $25000 (\frac{403}{400})^3 - A (\frac{403}{400})^2 - A (\frac{403}{400}) - A$

at the end of  $n$  years, i.e. 72 months:

$25000 (\frac{403}{400})^{72} = A [(\frac{403}{400})^{72} + (\frac{403}{400})^{71} + \dots + 1]$

$\therefore A = 25000 (\frac{403}{400})^{72} \div \frac{[(\frac{403}{400})^{72} - 1]}{(\frac{403}{400}) - 1}$

$\therefore A = \$450$  3 (\$450-64)

(ii) after the 24th repayment

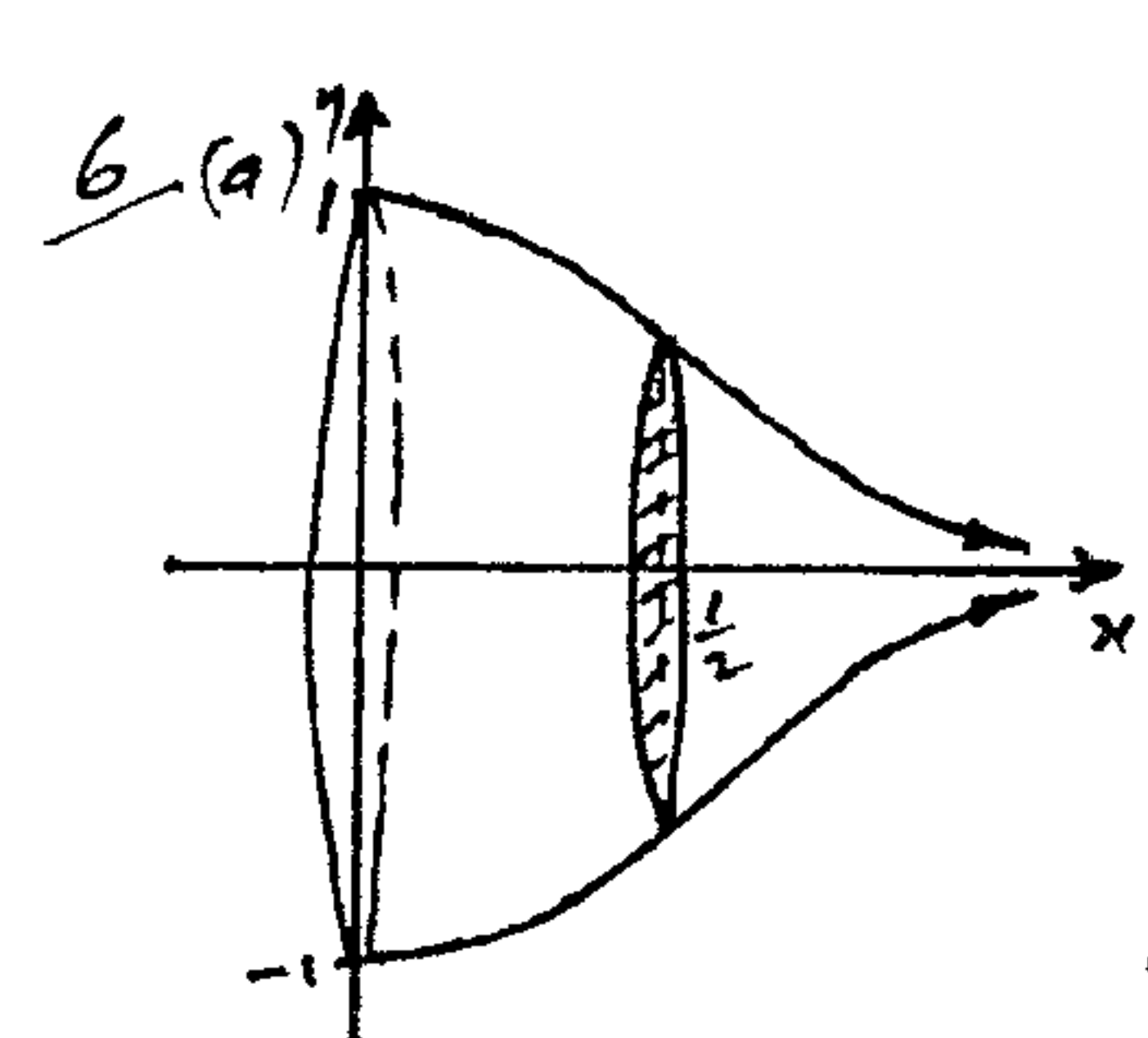
Let  $D = 25000 (\frac{403}{400})^{24} - 450 [(\frac{403}{400})^{24} - 1]$

$\therefore D = \$18,125.53$  2

(iii) Let  $P = \$18,126 - \$10,000 = \$8,126$

$\therefore 8,126 (\frac{403}{400})^{48} = A [1 + (\frac{403}{400}) + (\frac{403}{400})^2 + \dots + (\frac{403}{400})^{47}]$

$\therefore A = \$202.20$  2



(i)  $V = \pi \int_0^1 (\frac{1}{\sqrt{1+4x^2}})^2 dx$   
 $= \pi \int_0^1 \frac{1}{1+4x^2} dx$   
 $= \frac{\pi}{4} [\tan^{-1} 2x]_0^1$   
 $\therefore V = \frac{\pi}{8} \ln 3$  3

(ii) Now, let  
 $V = \frac{\pi}{2} [\tan^{-1} 2x]^x$   
As  $x \rightarrow \infty$ ,  $\tan^{-1} 2x \rightarrow \frac{\pi}{2}$   
 $\therefore V = \frac{\pi^2}{4} \ln 3$  2

(b) (i)  $\frac{d}{dx} (\frac{1}{(1+x)^2 \sqrt{x}}) = \frac{2}{(1+x)^2 \sqrt{x}} - \frac{1}{\sqrt{x}} \cdot \frac{1}{(1+x)^2}$

$= \frac{1}{(1+x)^2 \sqrt{x}} - \frac{1}{\sqrt{x}} \cdot \frac{1}{(1+x)^2}$   
 $= \frac{1}{(1+x)^2 \sqrt{x}} - \frac{1}{(1+x)^2 \sqrt{x}}$   
 $\therefore \frac{d}{dx} = 0$

(ii) Since  $\frac{d}{dx} = 0 \therefore f(x) = \text{constant}$

and since  $x > 0$ , let  $x = 1$

$\therefore f(x) = 2 \tan^{-1} 1 + \sin^{-1} 0 = \frac{\pi}{2}$

